Huawei TechArena 2025: BESS Energy Management System

Phase II Research Framework & Mathematical Model

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Based on Phase I Model and Literature Review

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Abstract

This document outlines the mathematical formulation for the Phase II challenge of the Huawei TechArena 2025. This model directly extends the deterministic Mixed-Integer Linear Program (MILP) developed in Phase I. Responding to the practical constraints of the competition (deterministic data, open-source solvers), this framework explicitly avoids stochastic optimization.

The Phase II model introduces two key extensions:

1. **Market Integration:** It incorporates the new **aFRR Energy Market**, co-optimizing energy bids alongside the Day-Ahead (DA) energy market and the FCR/aFRR capacity markets.
2. **Degradation Modeling:** It replaces the simple "Daily Cycle Limit" (Cst-5) from Phase I with a sophisticated, linearized **degradation cost function**. This cost, subtracted from the objective, is based on the piecewise-linear models for both cyclic aging (Xu et al., 2017) and calendar aging (Collath et al., 2023), allowing the model to trade off revenue against battery lifetime dynamically.

The resulting formulation remains a deterministic MILP, ensuring compatibility with open-source solvers while capturing the core economic and physical trade-offs of Phase II.

1 Introduction

The Phase I model established a strong foundation for co-optimizing BESS bids across three markets: DA Energy, FCR Capacity, and aFRR Capacity. The primary goal of Phase II is to enhance this model's realism and profitability by integrating two new factors: the aFRR Energy Market and the economic impact of battery degradation.

This document presents the revised mathematical formulation. It is designed to build directly upon the Phase I model, modifying existing constraints and adding new logic, rather than starting from a new, incompatible framework.

2 Phase II Model Derivation

We extend the Phase I MILP by modifying the objective function and relevant constraints.

2.1 Integrating the aFRR Energy Market

The aFRR Energy market (15-min granularity) operates similarly to the DA market as a price-taker energy market. We introduce new variables for bidding in this market and update all constraints that manage power and energy.

- **New Variables:** $p^{\text{pos}}_{aFRR,E}(t)$ (discharge/positive) and $p^{\text{neg}}_{aFRR,E}(t)$ (charge/negative) for energy bids in the aFRR energy market.
- **New Binaries:** $y^{\text{pos}}_{aFRR,E}(t)$ and $y^{\text{neg}}_{aFRR,E}(t)$ to manage minimum bids.

- **Objective Function:** A new profit term $\mathbb{P}^{aFRR\_E}$ is added:

$$\mathbb{P}^{aFRR\_E} = \sum_{t \in T} \left(\frac{P^{\text{pos}}_{aFRR,E}(t)}{1000}\, p^{\text{pos}}_{aFRR,E}(t) - \frac{P^{\text{neg}}_{aFRR,E}(t)}{1000}\, p^{\text{neg}}_{aFRR,E}(t) \right) \Delta t$$

This requires revising the core physical constraints to prevent conflicts. We define new **total** charge/discharge variables:

$$p^{\text{total}}_{\text{ch}}(t) = p_{\text{ch}}(t) + p^{\text{neg}}_{aFRR,E}(t) \quad \forall t$$
$$p^{\text{total}}_{\text{dis}}(t) = p_{\text{dis}}(t) + p^{\text{pos}}_{aFRR,E}(t) \quad \forall t$$

These totals are then used to update the SOC, power, and mutual exclusivity constraints from Phase I.

2.2 Integrating Battery Degradation as a Linear Cost

We replace the rigid Phase I constraint (Cst-5) $\sum p_{\text{dis}} \leq N_{\text{cycles}} E_{\text{nom}}$ with a flexible and more profitable **degradation cost function**, $C^{\text{Degradation}}$. This cost is subtracted from the objective function, allowing the optimizer to find the best balance between high-revenue, high-degradation actions and low-revenue, low-degradation actions.

Following the literature provided (Collath et al., 2023; Xu et al., 2017), we model this cost as the sum of calendar and cyclic aging:

$$C^{\text{Degradation}} = C^{\text{cal}} + C^{\text{cyc}}$$

2.2.1 Cyclic Aging Cost (C^{cyc})

Based on the methodology from Xu et al. (2017), we model cyclic aging as a **convex piecewise-linear cost of discharge**. The battery's energy capacity E_{nom} is divided into J segments (e.g., 10 segments of 10% SOC). Discharging from a shallower segment (e.g., 90-100% SOC) is "cheaper" than discharging from a deeper segment (e.g., 20-30% SOC).

- **New Variables:** $e_{\text{soc},j}(t)$ (energy in segment j), $p^{\text{ch}}_j(t)$ (charge to segment j), and $p^{\text{dis}}_j(t)$ (discharge from segment j).
- **Cost Function:** $C^{\text{cyc}} = \sum_{t \in T} \sum_{j \in J} \left(c^{\text{cost}}_j \cdot \frac{p^{\text{dis}}_j(t)}{\eta_{\text{dis}}} \cdot \Delta t \right)$
- **Logic:** The marginal cost c^{cost}_j increases as j increases (deeper discharge), e.g., $c_1 < c_2 < ... < c_J$. The optimizer will naturally prefer to discharge from the "cheapest" (shallowest) available segment first, perfectly modeling the non-linear cost of deep discharges in a linear framework.

2.2.2 Calendar Aging Cost (C^{cal})

Based on Collath et al. (2023), calendar aging is a non-linear function of the average State of Charge (SOC). We linearize this cost using Special Ordered Sets of Type 2 (SOS2), a standard MILP technique.

- **New Variables:** $\lambda_{t,i}$ (weighting variables for linearization).
- **Cost Function:** $C^{\text{cal}} = \sum_{t \in T} c^{\text{cal}}_{\text{cost}}(t)$
- **Logic:** The total SOC $e_{\text{soc}}(t)$ and its corresponding cost $c^{\text{cal}}_{\text{cost}}(t)$ are expressed as a convex combination of I pre-calculated (SOC, Cost) breakpoints from the literature's non-linear curve.

$$e_{\text{soc}}(t) = \sum_{i \in I} \lambda_{t,i} \cdot SOC^{\text{point}}_i \quad | \quad c^{\text{cal}}_{\text{cost}}(t) = \sum_{i \in I} \lambda_{t,i} \cdot Cost^{\text{point}}_i$$

$$\sum_{i \in I} \lambda_{t,i} = 1 \quad | \quad \{\lambda_{t,i}\} \text{ are SOS2 variables}$$

This formulation directly penalizes holding the battery at high SOC, as prioritized in the project description.

3 Phase II Mathematical Formulation

3.1 Nomenclature

3.1.1 Sets and Indices

T, t	Set and index for 15-minute time intervals $t \in \{1, ..., 35040\}$
B, b	Set and index for 4-hour ancillary service blocks $b \in \{1, ..., 2190\}$
J, j	Set and index for cycle degradation cost segments $j \in \{1, ..., J\}$
I, i	Set and index for calendar degradation cost breakpoints $i \in \{1, ..., I\}$

3.1.2 Parameters

Symbol	Description	Unit
$P_{DA}(t)$	Day-ahead electricity price	EUR/MWh
$P_{FCR}(b)$	FCR capacity price	EUR/MW
$P^{\text{pos}}_{aFRR}(b)$	Positive aFRR capacity price	EUR/MW
$P^{\text{neg}}_{aFRR}(b)$	Negative aFRR capacity price	EUR/MW
$P^{\text{pos}}_{aFRR,E}(t)$	**(New)** Positive aFRR energy price	EUR/MWh
$P^{\text{neg}}_{aFRR,E}(t)$	**(New)** Negative aFRR energy price	EUR/MWh
E_{nom}	Nominal energy capacity	kWh
$P^{\text{config}}_{\text{max}}$	Max charge/discharge power	kW
$\eta_{\text{ch}}, \eta_{\text{dis}}$	Charging/discharging efficiencies	-
$SOC_{\text{min}}, SOC_{\text{max}}$	Min/max SOC limits (fraction)	-
E^{seg}_j	**(New)** Max energy capacity of segment j	kWh
c^{cost}_j	**(New)** Marginal cyclic degradation cost for segment j	EUR/kWh
SOC^{point}_i	**(New)** SOC breakpoint i for calendar cost	kWh
$Cost^{\text{point}}_i$	**(New)** Calendar degradation cost at breakpoint i	EUR/hr

3.1.3 Decision Variables

Symbol	Description	Type
$p_{\text{ch}}(t), p_{\text{dis}}(t)$	DA charge/discharge power	Cont. ≥ 0
$p^{\text{pos}}_{aFRR,E}(t)$	**(New)** aFRR-E positive (discharge) power	Cont. ≥ 0
$p^{\text{neg}}_{aFRR,E}(t)$	**(New)** aFRR-E negative (charge) power	Cont. ≥ 0
$c_{fcr}(b)$	FCR capacity bid	Cont. ≥ 0
$c^{\text{pos}}_{aFRR}(b)$	Positive aFRR capacity bid	Cont. ≥ 0
$c^{\text{neg}}_{aFRR}(b)$	Negative aFRR capacity bid	Cont. ≥ 0
$p^{\text{total}}_{\text{ch}}(t), p^{\text{total}}_{\text{dis}}(t)$	**(New)** Total charge/discharge power	Cont. ≥ 0
$p^{\text{ch}}_j(t), p^{\text{dis}}_j(t)$	**(New)** Charge/discharge power for segment j	Cont. ≥ 0
$e_{\text{soc},j}(t)$	**(New)** Energy stored in segment j	Cont. ≥ 0
$e_{\text{soc}}(t)$	**(New)** Total energy stored in BESS	Cont. ≥ 0
$c^{\text{cal}}_{\text{cost}}(t)$	**(New)** Calendar cost at time t	Cont. ≥ 0
$\lambda_{t,i}$	**(New)** SOS2 variable for calendar cost	Cont. ≥ 0
$y_{\text{ch}}(t), y_{\text{dis}}(t)$	DA bid binaries	Binary
$y_{fcr}(b), y^{\text{pos}}_{aFRR}(b), ...$	Ancillary service bid binaries	Binary
$y^{\text{pos}}_{aFRR,E}(t), ...$	**(New)** aFRR-E bid binaries	Binary
$y^{\text{total}}_{\text{ch}}(t), y^{\text{total}}_{\text{dis}}(t)$	**(New)** Total operation binaries	Binary

3.2 Objective Function

The objective is to maximize the total profit, defined as market revenues minus degradation costs.

$$\max\ Z = \mathbb{P}^{DA} + \mathbb{P}^{ANCI} + \mathbb{P}^{aFRR\_E} - (C^{\text{cyc}} + C^{\text{cal}}) \tag{1}$$

Where:

$$\mathbb{P}^{DA} = \sum_{t\in T} \left(\frac{P_{DA}(t)}{1000}\, p_{\text{dis}}(t) - \frac{P_{DA}(t)}{1000}\, p_{\text{ch}}(t) \right) \Delta t \tag{2}$$

$$\mathbb{P}^{ANCI} = \sum_{b\in B} \left(P_{FCR}(b)\, c_{fcr}(b) + P^{\text{pos}}_{aFRR}(b)\, c^{\text{pos}}_{aFRR}(b) + P^{\text{neg}}_{aFRR}(b)\, c^{\text{neg}}_{aFRR}(b) \right) \tag{3}$$

$$\mathbb{P}^{aFRR\_E} = \sum_{t\in T} \left(\frac{P^{\text{pos}}_{aFRR,E}(t)}{1000}\, p^{\text{pos}}_{aFRR,E}(t) - \frac{P^{\text{neg}}_{aFRR,E}(t)}{1000}\, p^{\text{neg}}_{aFRR,E}(t) \right) \Delta t \tag{4}$$

$$C^{\text{cyc}} = \sum_{t\in T} \sum_{j\in J} \left(c^{\text{cost}}_j \cdot \frac{p^{\text{dis}}_j(t)}{\eta_{\text{dis}}} \cdot \Delta t \right) \tag{5}$$

$$C^{\text{cal}} = \sum_{t\in T} c^{\text{cal}}_{\text{cost}}(t) \cdot \Delta t \tag{6}$$

3.3 Constraints

3.3.1 Degradation-Aware SOC Dynamics (Replaces Cst-1)

$$e_{\text{soc},j}(t) = e_{\text{soc},j}(t-1) + \left(p^{\text{ch}}_j(t)\eta_{\text{ch}} - \frac{p^{\text{dis}}_j(t)}{\eta_{\text{dis}}} \right) \Delta t \qquad \forall t, \forall j \tag{7}$$

$$e_{\text{soc}}(t) = \sum_{j\in J} e_{\text{soc},j}(t) \qquad \forall t \tag{8}$$

$$p^{\text{total}}_{\text{ch}}(t) = \sum_{j\in J} p^{\text{ch}}_j(t) \qquad \forall t \tag{9}$$

$$p^{\text{total}}_{\text{dis}}(t) = \sum_{j\in J} p^{\text{dis}}_j(t) \qquad \forall t \tag{10}$$

$$p^{\text{total}}_{\text{ch}}(t) = p_{\text{ch}}(t) + p^{\text{neg}}_{aFRR,E}(t) \qquad \forall t \tag{11}$$

$$p^{\text{total}}_{\text{dis}}(t) = p_{\text{dis}}(t) + p^{\text{pos}}_{aFRR,E}(t) \qquad \forall t \tag{12}$$

3.3.2 SOC & Segment Limits (Replaces Cst-2)

$$SOC_{\min} E_{\text{nom}} \le e_{\text{soc}}(t) \le SOC_{\max} E_{\text{nom}} \qquad \forall t \tag{13}$$

$$0 \le e_{\text{soc},j}(t) \le E^{\text{seg}}_j \qquad \forall t, \forall j \tag{14}$$

3.3.3 Simultaneous Operation Prevention (Replaces Cst-3)

$$p^{\text{total}}_{\text{ch}}(t) \le y^{\text{total}}_{\text{ch}}(t) \cdot P^{\text{config}}_{\max} \qquad \forall t \tag{15}$$

$$p^{\text{total}}_{\text{dis}}(t) \le y^{\text{total}}_{\text{dis}}(t) \cdot P^{\text{config}}_{\max} \qquad \forall t \tag{16}$$

$$y^{\text{total}}_{\text{ch}}(t) + y^{\text{total}}_{\text{dis}}(t) \le 1 \qquad \forall t \tag{17}$$

3.3.4 Market Co-optimization Power Limits (Replaces Cst-4)

$$p_{\text{dis}}^{\text{total}}(t) + 1000\, c_{fcr}(b) + 1000\, c_{aFRR}^{\text{pos}}(b) \leq P_{\max}^{\text{config}} \qquad \forall b, t \in b \tag{18}$$

$$p_{\text{ch}}^{\text{total}}(t) + 1000\, c_{fcr}(b) + 1000\, c_{aFRR}^{\text{neg}}(b) \leq P_{\max}^{\text{config}} \qquad \forall b, t \in b \tag{19}$$

3.3.5 Ancillary Service Energy Reserve (Keeps Cst-6 Logic)

$$\frac{\left(1000\, c_{fcr}(b) + 1000\, c_{aFRR}^{\text{pos}}(b)\right) \tau}{\eta_{\text{dis}}} \leq e_{\text{soc}}(t) - SOC_{\min}\, E_{\text{nom}} \qquad \forall b, t \in b \tag{20}$$

$$\left(1000\, c_{fcr}(b) + 1000\, c_{aFRR}^{\text{neg}}(b)\right) \tau\, \eta_{\text{ch}} \leq SOC_{\max}\, E_{\text{nom}} - e_{\text{soc}}(t) \qquad \forall b, t \in b \tag{21}$$

(Where τ is the reserve duration, e.g., 0.25h)

3.3.6 Ancillary Capacity Market Exclusivity (Keeps Cst-7)

$$y_{fcr}(b) + y_{aFRR}^{\text{pos}}(b) + y_{aFRR}^{\text{neg}}(b) \leq 1 \qquad \forall b \tag{22}$$

3.3.7 Cross-Market Mutual Exclusivity (Replaces Cst-8)

$$y_{\text{ch}}^{\text{total}}(t) + y_{fcr}(b) + y_{aFRR}^{\text{pos}}(b) \leq 1 \qquad \forall b, t \in b \tag{23}$$

$$y_{\text{dis}}^{\text{total}}(t) + y_{fcr}(b) + y_{aFRR}^{\text{neg}}(b) \leq 1 \qquad \forall b, t \in b \tag{24}$$

3.3.8 Minimum Bids & Binary Logic (Updates Cst-9)

$$y_{\text{ch}}(t)\, \text{MinBid}_{da} \cdot 1000 \leq p_{\text{ch}}(t) \leq y_{\text{ch}}(t)\, P_{\max}^{\text{config}} \qquad \forall t \tag{25}$$

$$y_{\text{dis}}(t)\, \text{MinBid}_{da} \cdot 1000 \leq p_{\text{dis}}(t) \leq y_{\text{dis}}(t)\, P_{\max}^{\text{config}} \qquad \forall t \tag{26}$$

$$y_{aFRR,E}^{\text{pos}}(t)\, \text{MinBid}_{afrr\_e} \cdot 1000 \leq p_{aFRR,E}^{\text{pos}}(t) \leq y_{aFRR,E}^{\text{pos}}(t)\, P_{\max}^{\text{config}} \qquad \forall t \tag{27}$$

$$y_{aFRR,E}^{\text{neg}}(t)\, \text{MinBid}_{afrr\_e} \cdot 1000 \leq p_{aFRR,E}^{\text{neg}}(t) \leq y_{aFRR,E}^{\text{neg}}(t)\, P_{\max}^{\text{config}} \qquad \forall t \tag{28}$$

$$y_{\text{ch}}^{\text{total}}(t) \geq y_{\text{ch}}(t); \quad y_{\text{ch}}^{\text{total}}(t) \geq y_{aFRR,E}^{\text{neg}}(t) \qquad \forall t \tag{29}$$

$$y_{\text{dis}}^{\text{total}}(t) \geq y_{\text{dis}}(t); \quad y_{\text{dis}}^{\text{total}}(t) \geq y_{aFRR,E}^{\text{pos}}(t) \qquad \forall t \tag{30}$$

(Similar min/max bid constraints for c_{fcr}, c_{aFRR}^{pos}, c_{aFRR}^{neg} remain from Phase I)

3.3.9 Calendar Aging PWL Constraints (New)

$$e_{\text{soc}}(t) = \sum_{i \in I} \lambda_{t,i} \cdot SOC_i^{\text{point}} \qquad \forall t \tag{31}$$

$$c_{\text{cost}}^{\text{cal}}(t) = \sum_{i \in I} \lambda_{t,i} \cdot Cost_i^{\text{point}} \qquad \forall t \tag{32}$$

$$\sum_{i \in I} \lambda_{t,i} = 1 \qquad \forall t \tag{33}$$

$$\{\lambda_{t,i}\}_{i \in I} \text{ are SOS2 variables} \qquad \forall t \tag{34}$$

4 Conclusion and Next Steps

This Phase II formulation successfully integrates the aFRR energy market and a sophisticated, linearized degradation cost model into the Phase I MILP. It remains fully deterministic and is designed for open-source solvers.

Next Steps:

1. **Parameterization:** Populate the degradation cost parameters (c_j^{cost}, SOC_i^{point}, $Cost_i^{\text{point}}$) using the curves from the Collath et al. and Xu et al. papers, or from the competition's "ORC Battery Degradation Model" if details are available.

2. **Implementation:** Update the Phase I Pyomo model with these new variables and modified constraints.

3. **Solve & Analyze:** Run the optimization for all configurations and countries to generate the Phase II deliverables.

References

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